TryHackMe Writeup: Introductory Networking

Field	Details
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Program	DG Interns Hub, Week 1
Room	Introductory Networking
Room link	https://tryhackme.com/room/introtonetworking?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&sharerId=698d81bc7977c1741b32e1c8
Estimated time	20 min
Status	Completed (100%)

Objective

Understand basic networking theory and use some simple networking tools.

What I did, task by task

The OSI model

- A seven layer model that explains how data moves across a network: Physical, Data Link, Network, Transport, Session, Presentation, Application.

Packets, frames and encapsulation

- Data is wrapped with headers at each layer as it goes down (encapsulation) and unwrapped on the other side (decapsulation).

TCP and UDP

- TCP is reliable and connection based. It sets up a connection with a three way handshake: SYN, SYN-ACK, ACK.
- UDP is faster but does not guarantee delivery, so it suits streaming and DNS.

Networking tools

- ping checks if a host is reachable using ICMP.
- traceroute shows the path packets take and each hop on the way.
- whois gives registration details of a domain.
- dig queries DNS records for a domain.

Commands and tools used

Command / Tool	What it does
ping	Test whether a host is reachable
traceroute	Show the route and hops to a destination
whois	Look up domain registration details

dig	Query DNS records

Key takeaways

- Every network problem can be placed on a layer, which makes troubleshooting easier.
- TCP gives reliability and UDP gives speed.
- Ping, traceroute, whois and dig are basic tools I will use in the lab tasks.

Why this matters for a SOC role

Reading packet captures and network alerts needs a good grasp of layers, TCP handshakes and DNS. These tools are also used to investigate suspicious domains and IPs.

Proof of completion

